

ABSHealthcareLite

Product Master Book

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Table of Contents

- [ABSHealthcareLite Product Master Book](#)
 - [Purpose](#)
 - [1. Executive Summary](#)
 - [2. Product Vision](#)
 - [3. Product Objectives](#)
 - [4. Target Markets](#)
 - [5. Deployment Models](#)
 - [6. Product Editions](#)
 - [ABS Clinic Edition](#)
 - [ABS Diagnostic Edition](#)
 - [ABS Hospital Edition](#)
 - [ABS Enterprise Edition](#)
 - [ABS Telemedicine Edition](#)
 - [ABS AI Edition](#)
 - [7. Architectural Overview](#)
 - [8. Core Product Capabilities](#)
 - [9. Module Catalog Overview](#)
 - [10. Multi-Tenant SaaS Architecture](#)
 - [11. Security & Compliance](#)
 - [12. Localization Strategy](#)
 - [13. Mobile Strategy](#)
 - [14. API Strategy](#)
 - [15. AI Strategy](#)
 - [16. Product Roadmap](#)
 - [17. Competitive Advantages](#)
 - [18. Conclusion](#)
- [ABSHealthcareLite Product Master Book](#)
 - [Volume 2 – Functional Module Catalog](#)
 - [SECTION 1: EXECUTIVE SUMMARY](#)
 - [SECTION 2: MODULE CATALOG OVERVIEW](#)
 - [SECTION 3: DETAILED MODULE CATALOG](#)

- Module 01 – Company/Tenant Management
- Module 02 – User Management
- Module 03 – Role & Permission
- Module 04 – Audit Center
- Module 05 – Notification Center
- Module 06 – Localization
- Module 07 – Branch / Location
- Module 08 – Department Management
- Module 09 – Category Management
- Module 10 – Master Service Catalog
- Module 11 – Doctor Management
- Module 12 – Referral Doctor
- Module 13 – Ward / Cabin / Bed Setup
- Module 14 – Diagnostic Inventory
- Module 15 – Patient Registration & MPI
- Module 16 – Patient Profile & Ledger
- Module 17 – Appointment & Queue
- Module 18 – Doctor Worklist & Encounter
- Module 19 – Prescription Management
- Module 20 – Pharmacy Catalog
- Module 21 – Sample Collection
- Module 22 – Result Entry & Analyzer
- Module 23 – Result Verification
- Module 24 – Report Release & Delivery
- Module 25 – Radiology Workflow
- Module 26 – Medication Admin Record (MAR)
- Module 27 – Nursing Station & Tasks
- Module 28 – Bed & Occupancy Management
- Module 29 – Discharge Management
- Module 30 – Patient Portal & Self Service
- Module 31 – Appointment & Follow-Up
- Module 32 – Telemedicine & Virtual Care
- Module 40 – AI Prescription Capture

- SECTION 4: MODULE GROUP SUMMARIES
- SECTION 5: CROSS MODULE WORKFLOWS
 - Standard OPD to Lab Workflow
 - Telemedicine Workflow
 - AI Prescription Flow
- SECTION 6: MODULE DEPENDENCY HEATMAP
- SECTION 7: EDITION MAPPING
- SECTION 8: IMPLEMENTATION PRIORITY
- SECTION 9: PRODUCT MATURITY MAP
- SECTION 10: CONCLUSION
- ABSHealthcareLite Product Master Book
 - Volume 3 – Implementation, Deployment & Go-Live Guide
 - SECTION 1: EXECUTIVE SUMMARY
 - SECTION 2: IMPLEMENTATION METHODOLOGY
 - SECTION 3: DEPLOYMENT MODELS
 - SECTION 4: INFRASTRUCTURE REQUIREMENTS
 - SECTION 5: INITIAL SYSTEM SETUP
 - SECTION 6: SECURITY IMPLEMENTATION
 - SECTION 7: MASTER DATA IMPLEMENTATION
 - SECTION 8: PATIENT DATA MIGRATION
 - SECTION 9: MODULE IMPLEMENTATION SEQUENCE
 - SECTION 10: TRAINING FRAMEWORK
 - SECTION 11: TESTING FRAMEWORK
 - SECTION 12: GO-LIVE READINESS CHECKLIST
 - SECTION 13: GO-LIVE EXECUTION PLAN
 - SECTION 14: POST GO-LIVE SUPPORT
 - SECTION 15: BACKUP & DISASTER RECOVERY
 - SECTION 16: MULTI-TENANT SAAS OPERATIONS
 - SECTION 17: LOCALIZATION IMPLEMENTATION
 - SECTION 18: MOBILE & API READINESS
 - SECTION 19: AI IMPLEMENTATION STRATEGY
 - SECTION 20: HOSPITAL GO-LIVE MATURITY MODEL
 - SECTION 21: IMPLEMENTATION RISKS & MITIGATION

- SECTION 22: FINAL IMPLEMENTATION ROADMAP
- SECTION 23: APPENDIX A – IMPLEMENTATION CHECKLISTS
- SECTION 24: APPENDIX B – GO-LIVE SIGN-OFF TEMPLATES
- SECTION 25: CONCLUSION
- ABSHealthcareLite Product Master Book
 - Volume 4 – UI/UX Blueprint & Design System
 - SECTION 1: EXECUTIVE SUMMARY
 - SECTION 2: DESIGN PHILOSOPHY
 - SECTION 3: GLOBAL DESIGN SYSTEM
 - Color System
 - Typography
 - Spacing & Layout
 - Icons & Badges
 - Alerts & Forms
 - SECTION 4: MULTI-LANGUAGE UI STANDARDS
 - SECTION 5: RESPONSIVE DESIGN STANDARDS
 - SECTION 6: NAVIGATION STANDARDS
 - SECTION 7: STANDARD SCREEN TYPES
 - SECTION 8: FORM DESIGN STANDARDS
 - SECTION 9: DATA GRID STANDARDS
 - SECTION 10: DASHBOARD DESIGN STANDARDS
 - SECTION 11: WORKFLOW UX STANDARDS
 - SECTION 12: ACCESSIBILITY STANDARDS
 - SECTION 13: AUDIT & SECURITY UX
 - SECTION 14: NOTIFICATION UX
 - SECTION 15: MODULE SCREEN CATALOG
 - SECTION 16: SAMPLE DATA STANDARDS
 - SECTION 17: MASTER MENU ARCHITECTURE
 - SECTION 18: DESIGN TOKENS & FUTURE Figma READINESS
 - SECTION 19: MOCKUP DEVELOPMENT ROADMAP
 - SECTION 20: APPENDIX A - STANDARD PAGE TEMPLATES
 - 1. Standard List / Grid Template
 - 2. Standard Entry / Form Template
 - 3. Split-Screen Approval / AI Review Template

- [SECTION 21: APPENDIX B - ABSHEALTHCARELITE DESIGN COMMANDMENTS](#)
- [SECTION 22: CONCLUSION](#)

ABSHealthcareLite Product Master Book

Version: 1.0

Purpose

This document serves as the official executive, sales, investor, implementation, and product overview guide for ABSHealthcareLite. It synthesizes all documented modules and architectural blueprints into a single, professional enterprise reference, outlining the capabilities, vision, and strategic roadmap of the platform.

1. Executive Summary

ABSHealthcareLite is a next-generation, multi-tenant Software-as-a-Service (SaaS) Healthcare Enterprise Resource Planning (ERP) platform.

It exists to solve the fragmentation, inefficiency, and revenue leakage prevalent in modern healthcare facilities. By unifying outpatient (OPD), inpatient (IPD), diagnostics, pharmacy, and patient engagement into a single cohesive ecosystem, ABSHealthcareLite empowers healthcare providers to deliver superior clinical outcomes while optimizing operational efficiency.

Our **SaaS Vision** is to provide a highly scalable, cloud-native infrastructure where multiple hospitals, clinics, and diagnostic centers can operate securely on a unified codebase, strictly isolated by tenant boundaries.

Our **Global Healthcare Vision** is to transcend geographical and linguistic barriers, offering a fully localized, multilingual platform ready for deployment across diverse international markets, bringing world-class healthcare IT to emerging and established economies alike.

2. Product Vision

- **Hospital Digital Transformation:** Eradicating paper-based workflows through end-to-end digitization, from admission to discharge.

- **Diagnostic Automation:** Creating a seamless, barcode-driven pipeline from sample collection to pathologist verification and digital report release.
 - **Patient Engagement:** Empowering patients to take control of their health via self-service portals, digital health vaults, and transparent communication.
 - **Telemedicine:** Breaking geographic barriers by integrating virtual consultations directly into the core hospital workflow.
 - **AI Healthcare Automation:** Utilizing artificial intelligence to reduce administrative burden, capture unstructured data, and enhance clinical decision-making with human-in-the-loop safety.
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3. Product Objectives

- **Business Objectives:** Achieve rapid market penetration through scalable SaaS deployment, creating recurring revenue streams and high customer retention.
 - **Clinical Objectives:** Enhance patient safety through structured prescriptions, automated medication administration records (MAR), and strict clinical governance.
 - **Operational Objectives:** Maximize resource utilization (beds, doctors, machines) and minimize turnaround times (TAT) across all departments.
 - **Financial Objectives:** Prevent revenue leakage through integrated billing clearances, smart AI billing recommendations, and strict occupancy tracking.
 - **Technology Objectives:** Maintain a robust, future-proof architecture that supports current ASP.NET WebForms/SQL Server deployments while being fully prepared for MVC/Core, PostgreSQL/Supabase, and Mobile API migrations.
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4. Target Markets

Segments:

- Clinics & Polyclinics
- Diagnostic Centers & Pathology Labs
- General & Multi-Specialty Hospitals
- Medical Colleges & Teaching Hospitals
- Specialty Centers (e.g., Oncology, Maternity)
- Telemedicine & Virtual Care Providers

Regions:

- Bangladesh (Domestic Market)
 - Middle East (GCC)
 - Asia Pacific
 - Europe
 - Africa
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5. Deployment Models

- **Cloud SaaS:** Fully managed, multi-tenant cloud deployment for rapid onboarding and zero infrastructure overhead.
 - **Private Cloud:** Dedicated cloud instances for large enterprise networks requiring strict data sovereignty.
 - **On-Premise:** Local server deployments for hospitals in regions with unreliable internet connectivity or strict regulatory constraints.
 - **Hybrid:** Core data on-premise with cloud-bursting for patient portals and telemedicine.
 - **Future Multi-Region:** Distributed cloud architecture to comply with international data residency laws.
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6. Product Editions

ABS Clinic Edition

- **Included Modules:** Patient Registration, Appointments, Doctor Worklist, Prescription, Basic Billing.
- **Target Customers:** Solo practitioners, polyclinics, and small outpatient facilities.
- **Key Benefits:** Rapid deployment, low cost, streamlines daily patient flow.

ABS Diagnostic Edition

- **Included Modules:** Service Catalog, Sample Collection, Result Entry, Verification, Report Release.

- **Target Customers:** Standalone pathology labs and imaging centers.
- **Key Benefits:** Barcode-driven accuracy, rapid report TAT, and integrated quality control.

ABS Hospital Edition

- **Included Modules:** Clinic + Diagnostic + Ward/Bed Setup, Occupancy, Nursing Station, MAR, Discharge Management.
- **Target Customers:** Mid-to-large inpatient hospitals.
- **Key Benefits:** Complete patient lifecycle management, optimized bed turnover, and strict clinical governance.

ABS Enterprise Edition

- **Included Modules:** Hospital Edition + Multi-Branch Management, Advanced Audit, and Corporate Billing.
- **Target Customers:** Hospital chains and medical colleges.
- **Key Benefits:** Centralized control over decentralized operations, unified master data, and enterprise analytics.

ABS Telemedicine Edition

- **Included Modules:** Patient Portal, Telemedicine, Virtual Waiting Room, Remote Prescribing.
- **Target Customers:** Virtual care providers and hospitals expanding their digital footprint.
- **Key Benefits:** Opens new revenue streams and expands geographic patient reach.

ABS AI Edition

- **Included Modules:** Enterprise Edition + Module 40 (AI Prescription Capture & Smart Billing).
 - **Target Customers:** High-volume hospitals and diagnostic networks.
 - **Key Benefits:** Drastic reduction in manual data entry, optimized revenue capture, and advanced clinical decision support.
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7. Architectural Overview

ABSHealthcareLite is built on a strict layered architecture, ensuring separation of concerns and future scalability:

- **Platform Layer:** Manages multi-tenant boundaries, branches, and global configurations.
 - **Security Layer:** Enforces RBAC, authentication, and immutable audit trails.
 - **Master Data Layer:** Centralized dictionaries for departments, services, and catalogs.
 - **Patient Layer:** Master Patient Index (MPI) and demographic ledgers.
 - **Clinical Layer:** Doctor workspaces, encounters, and prescriptions.
 - **Diagnostic Layer:** LIS and RIS engines managing samples, results, and imaging.
 - **IPD Layer:** Inpatient logistics, nursing tasks, and bed occupancy.
 - **Finance Layer:** Revenue cycle, billing, and payment tracking.
 - **Patient Engagement Layer:** Portals, notifications, and self-service tools.
 - **Telemedicine Layer:** Virtual consultation and remote care infrastructure.
 - **AI Layer:** OCR, intelligent mapping, and machine learning pipelines.
 - **Analytics Layer:** Dashboards, KPIs, and enterprise reporting.
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8. Core Product Capabilities

- **Patient Management:** Unified identity tracking across all hospital touchpoints.
 - **Appointment Management:** Multi-channel scheduling, waitlists, and automated reminders.
 - **Clinical Documentation:** Structured EMR, progress notes, and clinical pathways.
 - **Laboratory Management:** End-to-end LIS with barcode tracking and multi-level verification.
 - **Radiology Management:** RIS workflow, scheduling, and structured dictation.
 - **Billing:** Integrated charge capture across all clinical and operational events.
 - **IPD:** Real-time bed boards, nursing handovers, and safe medication administration.
 - **Portal:** Secure patient access to records, appointments, and bills.
 - **Telemedicine:** Integrated video consults with remote prescribing.
 - **AI:** Automated data extraction from unstructured clinical documents.
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9. Module Catalog Overview

- **[01] Company/Tenant Management:** Foundation | Manages SaaS tenant boundaries and global settings.
- **[02] User Management:** Foundation | Controls staff identities and authentication.
- **[03] Role & Permission:** Foundation | Enforces granular, role-based access control.
- **[04] Audit Center:** Foundation | Centralized, immutable tracking of all system actions.
- **[05] Notification Center:** Foundation | Manages SMS, Email, and internal alerts.
- **[06] Localization:** Foundation | Drives the multilingual, RTL/LTR user interface.
- **[07] Branch/Location:** Master Data | Manages physical hospital locations within a tenant.
- **[08] Department Management:** Master Data | Configures clinical and operational departments.
- **[09] Category Management:** Master Data | Hierarchical grouping for services and inventory.
- **[10] Master Service Catalog:** Master Data | The central repository for all billable hospital services.
- **[11] Doctor Management:** Master Data | Manages provider profiles, specialties, and credentials.
- **[12] Referral Doctor:** Master Data | Tracks external referring physicians for commissions and reporting.
- **[13] Ward/Cabin/Bed Setup:** Master Data | Defines the physical inpatient infrastructure.
- **[14] Diagnostic Inventory:** Master Data | Manages lab reagents, consumables, and stock.
- **[15] Patient Registration & MPI:** Patient Journey | Creates the unified patient identity.
- **[16] Patient Profile Ledger:** Patient Journey | The 360-degree clinical and financial patient view.
- **[17] Appointment & Queue Mgmt:** Patient Journey | Basic scheduling and token generation.
- **[18] Doctor Worklist & Encounter:** Clinical | The physician's workspace for diagnosis and clinical notes.
- **[19] Prescription Management:** Clinical | Generates structured, multilingual medication plans.
- **[20] Pharmacy Medication Catalog:** Clinical | The master drug database driving prescriptions and MAR.

- [21] **Sample Collection:** Laboratory | Barcode generation and physical sample tracking.
- [22] **Result Entry & Analyzer:** Laboratory | Captures manual and machine-generated lab results.
- [23] **Result Verification:** Laboratory | Multi-level clinical approval and quality control.
- [24] **Report Release & Delivery:** Laboratory | Secure distribution of finalized diagnostic reports.
- [25] **Radiology Workflow:** Radiology | Manages imaging schedules, acquisition, and dictation.
- [26] **Medication Administration Record:** IPD | Tracks actual drug administration by nurses for patient safety.
- [27] **Nursing Station & Tasks:** IPD | The operational hub for vitals, handovers, and clinical tasks.
- [28] **Bed & Occupancy Management:** IPD | Real-time tracking of bed status, transfers, and housekeeping.
- [29] **Discharge Management:** IPD | Multi-department clearance and continuity of care planning.
- [30] **Patient Portal & Self Service:** Engagement | The patient-facing web and mobile gateway.
- [31] **Appointment & Follow-Up:** Engagement | Advanced scheduling, waitlists, and compliance tracking.
- [32] **Telemedicine & Virtual Consult:** Telemedicine | Integrated remote care and video consultations.
- [40] **AI Prescription Capture:** AI | Intelligent OCR and mapping of unstructured prescriptions to billing orders.

10. Multi-Tenant SaaS Architecture

ABSHealthcareLite is built from the ground up for the cloud. Every piece of data is tagged with a `CompanyId`, ensuring absolute **Data Isolation** between tenants. A single deployment can host hundreds of independent hospitals. Within a tenant, data is further segregated by **Branch** and **Department**. The architecture supports dynamic **Tenant Configuration**, allowing Hospital A to operate with different billing rules, languages, and approval workflows than Hospital B, all on the same codebase.

11. Security & Compliance

- **Authentication & Authorization:** Robust identity verification with strict RBAC.
 - **Audit:** Every transaction logs the actor, timestamp, and action. Historical data is never deleted, only soft-deleted (`IsActive = 0`).
 - **Privacy & Consent:** Built-in mechanisms to track patient consent for data sharing and telemedicine.
 - **AI & Telemedicine Governance:** Strict human-in-the-loop verification for AI decisions and secure, encrypted session management for virtual care.
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12. Localization Strategy

The platform is inherently borderless. It supports dynamic UI translation and data entry in:

- **English** (LTR)
 - **Bangla** (LTR)
 - **Hindi** (LTR)
 - **Arabic** (RTL)
 - **Urdu** (RTL) The architecture automatically mirrors the UI layout (Right-to-Left) based on the user's selected language, with no hardcoded strings in the application layer.
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13. Mobile Strategy

While currently optimized for responsive web browsers, the architecture is decoupled to support future native applications:

- **Patient App:** For portal access, telehealth, and self-service.
 - **Doctor App:** For schedule management, mobile EMR, and remote approvals.
 - **Future Nursing App:** For bedside vitals entry and Barcode Medication Administration (BCMA).
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14. API Strategy

The business logic is designed to be exposed via RESTful APIs. This ensures:

- **Mobile Readiness:** Seamless backend support for native apps.
 - **External Integration:** Connecting with third-party ERPs, accounting software, and external reference labs.
 - **FHIR Readiness:** Architectural alignment with global Fast Healthcare Interoperability Resources standards for health information exchange.
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15. AI Strategy

ABSHealthcareLite embraces Artificial Intelligence as an assistive tool, not a replacement for clinical judgment.

- **Module 40** leads this initiative by using OCR and NLP to digitize handwritten prescriptions and recommend billing packages.
 - **Clinical Safety Principles:** All AI extractions carry a confidence score. Low-confidence or high-risk items require mandatory human verification (Human-in-the-loop).
 - **Future Roadmap:** AI symptom triage, predictive bed occupancy forecasting, and readmission risk scoring.
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16. Product Roadmap

- **Current Phase:** Deployment of Modules 01-32 and 40, covering the complete clinical, diagnostic, and patient engagement lifecycle.
 - **Future Modules:** Comprehensive Finance & Accounting, HR & Payroll, Fixed Asset Management, and Advanced Pharmacy Procurement.
 - **Future Expansion:** Specialized clinical modules including Blood Bank, Operating Theater (OT) Management, Intensive Care Unit (ICU) flows, and Emergency Room (ER) triage.
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17. Competitive Advantages

- **Diagnostic Focus:** Unlike standard HIS platforms, ABSHealthcareLite contains a deeply integrated, enterprise-grade LIS and RIS.
 - **Integrated Telemedicine:** Virtual care is not a bolted-on third-party app; it is native to the ERP, sharing the same patient records and billing engine.
 - **AI Prescription Capture:** A unique market differentiator that turns unstructured paper into structured revenue.
 - **True Multilingual & RTL Support:** Ready for immediate deployment in the Middle East and South Asia without architectural rewrites.
 - **Multi-tenant Architecture:** Drastically lowers the Total Cost of Ownership (TCO) for hospital chains and SaaS operators.
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18. Conclusion

ABSHealthcareLite is more than a hospital management system; it is a comprehensive digital health ecosystem. By bridging the gap between robust inpatient logistics, advanced diagnostic automation, and modern patient engagement, it provides healthcare organizations with the tools they need to scale. Our long-term vision is to drive global healthcare interoperability, utilizing cloud architecture and AI to make high-quality healthcare administration accessible, efficient, and secure anywhere in the world.

ABSHealthcareLite Product Master Book

Volume 2 – Functional Module Catalog

SECTION 1: EXECUTIVE SUMMARY

The **ABSHealthcareLite Product Master Book Volume 2 – Functional Module Catalog** provides a comprehensive, functional breakdown of all 33 documented modules within the ABSHealthcareLite ecosystem.

Designed for Product Managers, Sales Teams, Implementation Specialists, and Healthcare Executives, this document translates the technical architecture into business capabilities. It works in tandem with **Volume 1 (Product Master Book)** and the **System Architecture Blueprints (01-03)** to provide a complete understanding of what the system does, who uses it, and how it is deployed.

This catalog reinforces our commitment to a Multi-tenant SaaS architecture, localized for global markets (English, Bangla, Arabic, Urdu, Hindi), and built on a future-proof technology stack (SQL Server to PostgreSQL, WebForms to MVC/Core/Mobile APIs).

SECTION 2: MODULE CATALOG OVERVIEW

Module	Name	Category	Criticality	Status
01	Company/Tenant Management	Foundation	Foundation	Documented
02	User Management	Security	Foundation	Documented
03	Role & Permission	Security	Foundation	Documented
04	Audit Center	Foundation	Foundation	Documented
05	Notification Center	Foundation	Foundation	Documented
06	Localization	Foundation	Foundation	Documented
07	Branch / Location	Master Data	Foundation	Documented
08	Department Management	Master Data	Foundation	Documented
09	Category Management	Master Data	Foundation	Documented
10	Master Service Catalog	Master Data	Foundation	Documented
11	Doctor Management	Master Data	Foundation	Documented
12	Referral Doctor	Master Data	Foundation	Documented
13	Ward / Cabin / Bed Setup	Master Data	Foundation	Documented
14	Diagnostic Inventory	Master Data	Foundation	Documented
15	Patient Registration & MPI	Patient	Core	Documented
16	Patient Profile & Ledger	Patient	Core	Documented
17	Appointment & Queue	Patient	Core	Documented
18	Doctor Worklist & Encounter	Clinical	Core	Documented
19	Prescription Management	Clinical	Core	Documented
20	Pharmacy Catalog	Clinical	Core	Documented
21	Sample Collection	Laboratory	Core	Documented

22	Result Entry & Analyzer	Laboratory	Advanced	Documented
23	Result Verification	Laboratory	Advanced	Documented
24	Report Release & Delivery	Laboratory	Advanced	Documented
25	Radiology Workflow	Radiology	Advanced	Documented
26	Medication Admin Record (MAR)	Nursing	Enterprise	Documented
27	Nursing Station & Tasks	Nursing	Enterprise	Documented
28	Bed & Occupancy Management	IPD	Enterprise	Documented
29	Discharge Management	IPD	Enterprise	Documented
30	Patient Portal & Self Service	Engagement	Advanced	Documented
31	Appointment & Follow-Up	Engagement	Advanced	Documented
32	Telemedicine & Virtual Care	Telemedicine	Enterprise	Documented
40	AI Prescription Capture	AI	AI	Documented

SECTION 3: DETAILED MODULE CATALOG

Module 01 – Company/Tenant Management

1. **Purpose:** Establishes the SaaS multi-tenant boundaries.
2. **Business Objectives:** Isolate data per hospital/clinic to ensure privacy and enable scalable cloud hosting.
3. **Primary Actors:** Host Admin.
4. **Secondary Actors:** Tenant Admin.
5. **Key Features:** Tenant onboarding, global settings, subscription limits, data isolation.
6. **Major Screens:** Host Dashboard, Tenant Setup.
7. **Major Reports:** Tenant List, Subscription Status.

8. **Major Database Entities:** Company , TenantConfig .
9. **Dependencies:** None.
10. **Used By:** All Modules.
11. **Security Considerations:** Absolute isolation via CompanyId .
12. **Audit Requirements:** Track tenant creation and status changes.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High (Provisioning APIs).
15. **AI Readiness:** Low.
16. **Future Expansion:** Multi-region data residency routing.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 02 – User Management

1. **Purpose:** Manages staff identities and authentication.
2. **Business Objectives:** Secure system access and track employee activity.
3. **Primary Actors:** Tenant Admin, HR.
4. **Secondary Actors:** All Staff.
5. **Key Features:** User creation, credential management, status toggling.
6. **Major Screens:** User Directory, User Profile.
7. **Major Reports:** Active Users, Login History.
8. **Major Database Entities:** User , UserLoginHistory .
9. **Dependencies:** Mod 01, 03.
10. **Used By:** All Modules.
11. **Security Considerations:** Password hashing, session timeouts.
12. **Audit Requirements:** Track user creation, suspension, and failed logins.
13. **Mobile Readiness:** High (Mobile login).
14. **API Readiness:** High (Auth endpoints).
15. **AI Readiness:** Low.
16. **Future Expansion:** SSO, Active Directory integration, Biometrics.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 03 – Role & Permission

1. **Purpose:** Enforces Role-Based Access Control (RBAC).
2. **Business Objectives:** Ensure users only access data necessary for their job.
3. **Primary Actors:** Tenant Admin.
4. **Secondary Actors:** None.
5. **Key Features:** Role creation, granular permission matrix, menu visibility control.
6. **Major Screens:** Role Manager, Permission Matrix.
7. **Major Reports:** Role Assignment Report.
8. **Major Database Entities:** Role , Permission , RolePermission .
9. **Dependencies:** Mod 01.
10. **Used By:** Mod 02.
11. **Security Considerations:** Prevents unauthorized API/URL access.
12. **Audit Requirements:** Track permission changes.
13. **Mobile Readiness:** Medium.
14. **API Readiness:** High (Middleware validation).
15. **AI Readiness:** Low.
16. **Future Expansion:** Attribute-based access control (ABAC).
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 04 – Audit Center

1. **Purpose:** Centralized tracking of all system data changes.
2. **Business Objectives:** Ensure medico-legal compliance and operational transparency.
3. **Primary Actors:** System Admin, Auditor.
4. **Secondary Actors:** Tenant Admin.
5. **Key Features:** Record-level tracking (Created, Modified, Deleted).
6. **Major Screens:** Global Audit Viewer.
7. **Major Reports:** System Audit Trail, Security Exceptions.
8. **Major Database Entities:** AuditLog .
9. **Dependencies:** Mod 01, 02.
10. **Used By:** All Modules.
11. **Security Considerations:** Audit logs must be immutable.

12. **Audit Requirements:** Self-auditing.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** Blockchain-backed audit trails.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 05 – Notification Center

1. **Purpose:** Manages outbound and internal communications.
2. **Business Objectives:** Improve patient engagement and staff responsiveness.
3. **Primary Actors:** System Admin.
4. **Secondary Actors:** All Users, Patients.
5. **Key Features:** SMS, Email, WhatsApp, and internal push notifications.
6. **Major Screens:** Notification Templates, Dispatch Log.
7. **Major Reports:** Delivery Success/Failure Report.
8. **Major Database Entities:** NotificationTemplate , NotificationLog .
9. **Dependencies:** Mod 01.
10. **Used By:** Mod 17, 23, 27, 30, 31.
11. **Security Considerations:** PHI masking in external messages.
12. **Audit Requirements:** Track message dispatch and delivery status.
13. **Mobile Readiness:** High (Push notifications).
14. **API Readiness:** High (Webhook integrations).
15. **AI Readiness:** Medium (Smart delivery timing).
16. **Future Expansion:** Conversational AI chatbots.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 06 – Localization

1. **Purpose:** Drives the multilingual, RTL/LTR user interface.
2. **Business Objectives:** Enable global deployment across diverse linguistic regions.
3. **Primary Actors:** Host Admin, Translators.
4. **Secondary Actors:** All Users.

5. **Key Features:** Dynamic language switching, RTL mirroring, resource file management.
6. **Major Screens:** Language Setup, Translation Dictionary.
7. **Major Reports:** Translation Coverage Report.
8. **Major Database Entities:** Language , TranslationKey .
9. **Dependencies:** Mod 01.
10. **Used By:** All Modules.
11. **Security Considerations:** Sanitization of translated inputs.
12. **Audit Requirements:** Track dictionary updates.
13. **Mobile Readiness:** High.
14. **API Readiness:** High.
15. **AI Readiness:** Medium (Auto-translation).
16. **Future Expansion:** Real-time clinical translation.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 07 – Branch / Location

1. **Purpose:** Manages physical hospital locations within a tenant.
2. **Business Objectives:** Support multi-facility hospital chains under one corporate umbrella.
3. **Primary Actors:** Tenant Admin.
4. **Secondary Actors:** None.
5. **Key Features:** Branch creation, address tracking, facility settings.
6. **Major Screens:** Branch Setup.
7. **Major Reports:** Branch List.
8. **Major Database Entities:** Branch .
9. **Dependencies:** Mod 01.
10. **Used By:** Mod 08, 13, 15, 28.
11. **Security Considerations:** Branch-level data filtering.
12. **Audit Requirements:** Track branch creation/closure.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** Geofencing.

- 17. **Criticality Level:** Foundation.
- 18. **Deployment Edition:** Hospital, Enterprise, AI Editions.

Module 08 – Department Management

- 1. **Purpose:** Configures clinical and operational departments.
- 2. **Business Objectives:** Organize staff, services, and workflows logically.
- 3. **Primary Actors:** Tenant Admin.
- 4. **Secondary Actors:** None.
- 5. **Key Features:** Clinical vs. Non-clinical classification, branch mapping.
- 6. **Major Screens:** Department Setup.
- 7. **Major Reports:** Department List.
- 8. **Major Database Entities:** Department .
- 9. **Dependencies:** Mod 01, 07.
- 10. **Used By:** Mod 10, 11, 17, 21, 28.
- 11. **Security Considerations:** Department-level access restrictions.
- 12. **Audit Requirements:** Track department changes.
- 13. **Mobile Readiness:** Low.
- 14. **API Readiness:** High.
- 15. **AI Readiness:** Low.
- 16. **Future Expansion:** Departmental budget mapping.
- 17. **Criticality Level:** Foundation.
- 18. **Deployment Edition:** All Editions.

Module 09 – Category Management

- 1. **Purpose:** Hierarchical grouping for services and inventory.
- 2. **Business Objectives:** Standardize reporting and billing classifications.
- 3. **Primary Actors:** Tenant Admin.
- 4. **Secondary Actors:** None.
- 5. **Key Features:** Parent/Child categories, tax mappings.
- 6. **Major Screens:** Category Tree Setup.
- 7. **Major Reports:** Category Hierarchy.
- 8. **Major Database Entities:** Category .
- 9. **Dependencies:** Mod 01.

10. **Used By:** Mod 10, 14, 20.
11. **Security Considerations:** None specific.
12. **Audit Requirements:** Track category changes.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** Global standard mappings (UNSPSC).
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 10 – Master Service Catalog

1. **Purpose:** The central repository for all billable hospital services.
2. **Business Objectives:** Ensure accurate billing and standardized clinical ordering.
3. **Primary Actors:** Billing Admin, Lab Admin.
4. **Secondary Actors:** Doctors.
5. **Key Features:** Service pricing, department mapping, lab test definitions.
6. **Major Screens:** Service Master Setup, Pricing Matrix.
7. **Major Reports:** Service Price List.
8. **Major Database Entities:** ServiceMaster , ServicePrice .
9. **Dependencies:** Mod 01, 08, 09.
10. **Used By:** Mod 18, 21, 22, 40.
11. **Security Considerations:** Price modification tracking.
12. **Audit Requirements:** Strict audit on price changes.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High.
15. **AI Readiness:** High (Target for Module 40 NLP mapping).
16. **Future Expansion:** Dynamic pricing algorithms.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 11 – Doctor Management

1. **Purpose:** Manages provider profiles, specialties, and credentials.
2. **Business Objectives:** Track physician availability, commissions, and clinical authority.

3. **Primary Actors:** HR, Tenant Admin.
4. **Secondary Actors:** Appointment Desk.
5. **Key Features:** Doctor profiles, specialty mapping, schedule templates.
6. **Major Screens:** Doctor Profile, Roster Setup.
7. **Major Reports:** Doctor Directory.
8. **Major Database Entities:** DoctorMaster .
9. **Dependencies:** Mod 01, 02, 08.
10. **Used By:** Mod 17, 18, 31, 32.
11. **Security Considerations:** Credential verification tracking.
12. **Audit Requirements:** Profile updates.
13. **Mobile Readiness:** High (Doctor App profile).
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** National medical council API integration.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** All Editions.

Module 12 – Referral Doctor

1. **Purpose:** Tracks external referring physicians.
2. **Business Objectives:** Manage B2B relationships and referral commissions.
3. **Primary Actors:** Marketing/CRM, Billing.
4. **Secondary Actors:** Reception.
5. **Key Features:** External doctor profiles, clinic mapping.
6. **Major Screens:** Referral Doctor Setup.
7. **Major Reports:** Referral Volume Report.
8. **Major Database Entities:** ReferralDoctor .
9. **Dependencies:** Mod 01.
10. **Used By:** Mod 15, 24.
11. **Security Considerations:** Financial commission visibility.
12. **Audit Requirements:** Track profile creation.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High.
15. **AI Readiness:** Low.

- 16. **Future Expansion:** Automated commission payouts.
- 17. **Criticality Level:** Foundation.
- 18. **Deployment Edition:** Diagnostic, Hospital, Enterprise Editions.

Module 13 – Ward / Cabin / Bed Setup

- 1. **Purpose:** Defines the physical inpatient infrastructure.
- 2. **Business Objectives:** Map the hospital's physical capacity for occupancy tracking.
- 3. **Primary Actors:** Tenant Admin, Facilities Manager.
- 4. **Secondary Actors:** None.
- 5. **Key Features:** Building/Floor/Ward/Room/Bed hierarchy creation.
- 6. **Major Screens:** Location Hierarchy Builder.
- 7. **Major Reports:** Bed Inventory List.
- 8. **Major Database Entities:** HospitalWard , HospitalRoom , HospitalBed .
- 9. **Dependencies:** Mod 01, 07.
- 10. **Used By:** Mod 28.
- 11. **Security Considerations:** None specific.
- 12. **Audit Requirements:** Track capacity changes.
- 13. **Mobile Readiness:** Low.
- 14. **API Readiness:** High.
- 15. **AI Readiness:** Low.
- 16. **Future Expansion:** 3D Digital Twin mapping.
- 17. **Criticality Level:** Foundation.
- 18. **Deployment Edition:** Hospital, Enterprise, AI Editions.

Module 14 – Diagnostic Inventory

- 1. **Purpose:** Manages lab reagents, consumables, and stock.
- 2. **Business Objectives:** Ensure labs never run out of critical testing supplies.
- 3. **Primary Actors:** Lab Manager, Store Keeper.
- 4. **Secondary Actors:** None.
- 5. **Key Features:** Item master, batch/expiry tracking, supplier mapping.
- 6. **Major Screens:** Item Setup, Stock Ledger.
- 7. **Major Reports:** Low Stock Alert, Expiry Report.
- 8. **Major Database Entities:** DiagnosticItem , StockLedger .

9. **Dependencies:** Mod 01, 09.
10. **Used By:** Mod 21, 22.
11. **Security Considerations:** Inventory value protection.
12. **Audit Requirements:** Track stock adjustments.
13. **Mobile Readiness:** Medium (Barcode scanning).
14. **API Readiness:** High.
15. **AI Readiness:** Medium (Predictive ordering).
16. **Future Expansion:** Automated supplier PO generation.
17. **Criticality Level:** Foundation.
18. **Deployment Edition:** Diagnostic, Hospital, Enterprise Editions.

Module 15 – Patient Registration & MPI

1. **Purpose:** Creates the unified patient identity.
2. **Business Objectives:** Prevent duplicate records and establish the Master Patient Index (MPI).
3. **Primary Actors:** Receptionist.
4. **Secondary Actors:** Patient (via Portal).
5. **Key Features:** Demographic capture, duplicate detection, ID card generation.
6. **Major Screens:** Registration Desk, Patient Search.
7. **Major Reports:** Daily Registration Report.
8. **Major Database Entities:** Patient .
9. **Dependencies:** Mod 01, 06, 07.
10. **Used By:** Mod 16, 17, 28, 30.
11. **Security Considerations:** Strict PHI protection.
12. **Audit Requirements:** Track demographic changes.
13. **Mobile Readiness:** High (Self-registration).
14. **API Readiness:** High.
15. **AI Readiness:** Medium (Face matching for deduplication).
16. **Future Expansion:** National ID integration.
17. **Criticality Level:** Core.
18. **Deployment Edition:** All Editions.

Module 16 – Patient Profile & Ledger

1. **Purpose:** The 360-degree clinical and financial patient view.
2. **Business Objectives:** Provide a single source of truth for a patient's history.
3. **Primary Actors:** Doctors, Nurses, Billing.
4. **Secondary Actors:** None.
5. **Key Features:** Timeline view, allergy tracking, consolidated ledger.
6. **Major Screens:** Patient 360 Dashboard.
7. **Major Reports:** Patient History Summary.
8. **Major Database Entities:** PatientProfile , PatientAllergy .
9. **Dependencies:** Mod 15.
10. **Used By:** Mod 18, 27, 29, 30.
11. **Security Considerations:** High PHI visibility restrictions.
12. **Audit Requirements:** Track profile views (Break-the-glass).
13. **Mobile Readiness:** High.
14. **API Readiness:** High.
15. **AI Readiness:** High (Risk profiling).
16. **Future Expansion:** Longitudinal health graphs.
17. **Criticality Level:** Core.
18. **Deployment Edition:** All Editions.

Module 17 – Appointment & Queue

1. **Purpose:** Basic scheduling and token generation.
2. **Business Objectives:** Manage daily outpatient flow.
3. **Primary Actors:** Front Desk.
4. **Secondary Actors:** Doctors.
5. **Key Features:** Slot booking, token printing, check-in.
6. **Major Screens:** Daily Roster, Token Queue.
7. **Major Reports:** Daily Appointment Register.
8. **Major Database Entities:** Appointment .
9. **Dependencies:** Mod 11, 15.
10. **Used By:** Mod 18, 31.
11. **Security Considerations:** None specific.

12. **Audit Requirements:** Track cancellations.
13. **Mobile Readiness:** High.
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** Replaced/Enhanced by Module 31.
17. **Criticality Level:** Core.
18. **Deployment Edition:** Clinic, Diagnostic Editions.

Module 18 – Doctor Worklist & Encounter

1. **Purpose:** The physician's primary clinical workspace.
2. **Business Objectives:** Digitally capture complaints, diagnoses, and orders.
3. **Primary Actors:** Doctors.
4. **Secondary Actors:** Nurses (Triage).
5. **Key Features:** Vitals review, clinical notes, ICD-10 coding, order entry.
6. **Major Screens:** Doctor Dashboard, Encounter Workspace.
7. **Major Reports:** Outpatient Morbidity Report.
8. **Major Database Entities:** Encounter , ClinicalNote .
9. **Dependencies:** Mod 11, 15, 17.
10. **Used By:** Mod 19, 21, 25, 29.
11. **Security Considerations:** Doctor-only edit rights.
12. **Audit Requirements:** Immutable clinical notes.
13. **Mobile Readiness:** High (Tablet EMR).
14. **API Readiness:** High.
15. **AI Readiness:** High (Voice-to-text notes).
16. **Future Expansion:** Clinical Decision Support (CDS).
17. **Criticality Level:** Core.
18. **Deployment Edition:** Clinic, Hospital, Enterprise, Telemedicine, AI Editions.

Module 19 – Prescription Management

1. **Purpose:** Generates structured, multilingual medication plans.
2. **Business Objectives:** Eliminate handwriting errors and standardize prescribing.
3. **Primary Actors:** Doctors.
4. **Secondary Actors:** Pharmacists.

5. **Key Features:** Drug search, dosage templates, multilingual printouts.
6. **Major Screens:** ePrescription Pad.
7. **Major Reports:** Prescribing Patterns Report.
8. **Major Database Entities:** Prescription , PrescriptionItem .
9. **Dependencies:** Mod 18, 20.
10. **Used By:** Mod 26, 30, 32.
11. **Security Considerations:** Prevents unauthorized prescription generation.
12. **Audit Requirements:** Track prescription edits.
13. **Mobile Readiness:** High.
14. **API Readiness:** High.
15. **AI Readiness:** Medium (Drug interaction alerts).
16. **Future Expansion:** National ePrescription gateway integration.
17. **Criticality Level:** Core.
18. **Deployment Edition:** Clinic, Hospital, Enterprise, Telemedicine, AI Editions.

Module 20 – Pharmacy Catalog

1. **Purpose:** The master drug database driving prescriptions and MAR.
2. **Business Objectives:** Standardize generic and brand names across the hospital.
3. **Primary Actors:** Chief Pharmacist.
4. **Secondary Actors:** Doctors, Nurses.
5. **Key Features:** Generic/Brand mapping, route/dosage forms, high-alert flags.
6. **Major Screens:** Drug Master Setup.
7. **Major Reports:** Formulary List.
8. **Major Database Entities:** PharmacyItem , GenericMaster .
9. **Dependencies:** Mod 01, 09.
10. **Used By:** Mod 19, 26, 29.
11. **Security Considerations:** Formulary control.
12. **Audit Requirements:** Track drug additions/removals.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** Global drug database API sync.
17. **Criticality Level:** Core.

18. **Deployment Edition:** Clinic, Hospital, Enterprise, AI Editions.

Module 21 – Sample Collection

1. **Purpose:** Barcode generation and physical sample tracking.
2. **Business Objectives:** Eliminate pre-analytical lab errors and lost samples.
3. **Primary Actors:** Phlebotomist.
4. **Secondary Actors:** Lab Reception.
5. **Key Features:** Smart tube grouping, barcode printing, chain of custody.
6. **Major Screens:** Phlebotomy Dashboard, Collection Screen.
7. **Major Reports:** Pending Collection Report.
8. **Major Database Entities:** SampleCollection , SampleBarcode .
9. **Dependencies:** Mod 10, 15, 18.
10. **Used By:** Mod 22.
11. **Security Considerations:** Patient verification before draw.
12. **Audit Requirements:** Track collection timestamps.
13. **Mobile Readiness:** High (Mobile barcode scanning).
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** Automated tube sorting integration.
17. **Criticality Level:** Core.
18. **Deployment Edition:** Diagnostic, Hospital, Enterprise, AI Editions.

Module 22 – Result Entry & Analyzer

1. **Purpose:** Captures manual and machine-generated lab results.
2. **Business Objectives:** Rapid, accurate diagnostic data entry.
3. **Primary Actors:** Lab Technologist.
4. **Secondary Actors:** None.
5. **Key Features:** Reference range auto-flagging, critical value alerts, analyzer queue.
6. **Major Screens:** Result Entry Grid, Analyzer Queue.
7. **Major Reports:** Pending Result Report.
8. **Major Database Entities:** LabResult , AnalyzerMapping .
9. **Dependencies:** Mod 21, 10.
10. **Used By:** Mod 23.

11. **Security Considerations:** Result tampering prevention.
12. **Audit Requirements:** Track manual vs. machine entry.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High (HL7/ASTM).
15. **AI Readiness:** Medium (Delta check anomaly detection).
16. **Future Expansion:** Bidirectional analyzer interfaces.
17. **Criticality Level:** Advanced.
18. **Deployment Edition:** Diagnostic, Hospital, Enterprise, AI Editions.

Module 23 – Result Verification

1. **Purpose:** Multi-level clinical approval and quality control.
2. **Business Objectives:** Ensure diagnostic accuracy and medico-legal compliance.
3. **Primary Actors:** Pathologist, Lab Supervisor.
4. **Secondary Actors:** None.
5. **Key Features:** Approval matrices, critical value notification tracking, correction workflow.
6. **Major Screens:** Verification Dashboard, Approval Queue.
7. **Major Reports:** Verification TAT Report.
8. **Major Database Entities:** ResultVerification , ResultApproval .
9. **Dependencies:** Mod 22, 02, 03.
10. **Used By:** Mod 24, 29.
11. **Security Considerations:** Digital signature tracking.
12. **Audit Requirements:** Strict audit of report amendments.
13. **Mobile Readiness:** High (Pathologist mobile approval).
14. **API Readiness:** High.
15. **AI Readiness:** High (Auto-verification rules).
16. **Future Expansion:** Peer review workflows.
17. **Criticality Level:** Advanced.
18. **Deployment Edition:** Diagnostic, Hospital, Enterprise, AI Editions.

Module 24 – Report Release & Delivery

1. **Purpose:** Secure distribution of finalized diagnostic reports.
2. **Business Objectives:** Deliver reports quickly while enforcing billing holds.

3. **Primary Actors:** Report Desk, Patients.
4. **Secondary Actors:** None.
5. **Key Features:** Release blocking matrix, QR authenticity, PDF generation.
6. **Major Screens:** Release Queue, Reprint Screen.
7. **Major Reports:** Delivery Report, Reprint Audit.
8. **Major Database Entities:** ReportRelease , ReportDelivery .
9. **Dependencies:** Mod 23, 15.
10. **Used By:** Mod 30.
11. **Security Considerations:** Prevent unauthorized downloads.
12. **Audit Requirements:** Track every print and download.
13. **Mobile Readiness:** High.
14. **API Readiness:** High.
15. **AI Readiness:** Low.
16. **Future Expansion:** Blockchain document verification.
17. **Criticality Level:** Advanced.
18. **Deployment Edition:** Diagnostic, Hospital, Enterprise, AI Editions.

Module 25 – Radiology Workflow

1. **Purpose:** Manages imaging schedules, acquisition, and dictation.
2. **Business Objectives:** Optimize expensive machine utilization and radiologist TAT.
3. **Primary Actors:** Radiologist, Rad Tech.
4. **Secondary Actors:** Reception.
5. **Key Features:** Modality scheduling, structured reporting templates, critical findings.
6. **Major Screens:** Rad Dashboard, Structured Reporting Screen.
7. **Major Reports:** Machine Utilization Report.
8. **Major Database Entities:** RadiologyStudy , RadiologyReport .
9. **Dependencies:** Mod 10, 15, 18.
10. **Used By:** Mod 24, 30.
11. **Security Considerations:** DICOM metadata protection.
12. **Audit Requirements:** Track acquisition times.
13. **Mobile Readiness:** Medium.
14. **API Readiness:** High (DICOM/PACS).
15. **AI Readiness:** High (AI image pre-screening).

- 16. **Future Expansion:** Full VNA (Vendor Neutral Archive) integration.
- 17. **Criticality Level:** Advanced.
- 18. **Deployment Edition:** Diagnostic, Hospital, Enterprise, AI Editions.

Module 26 – Medication Admin Record (MAR)

- 1. **Purpose:** Tracks actual drug administration by nurses for patient safety.
- 2. **Business Objectives:** Enforce the "Rights of Medication" and prevent errors.
- 3. **Primary Actors:** Nurses.
- 4. **Secondary Actors:** Doctors, Pharmacists.
- 5. **Key Features:** Scheduled/PRN tracking, missed dose governance, adverse event logging.
- 6. **Major Screens:** MAR Dashboard, Administration Screen.
- 7. **Major Reports:** Missed Dose Report.
- 8. **Major Database Entities:** MedicationAdministrationRecord .
- 9. **Dependencies:** Mod 19, 20, 27, 28.
- 10. **Used By:** Mod 29.
- 11. **Security Considerations:** Witness signatures for narcotics.
- 12. **Audit Requirements:** Immutable administration logs.
- 13. **Mobile Readiness:** High (Bedside tablet).
- 14. **API Readiness:** High.
- 15. **AI Readiness:** Medium (Adherence analytics).
- 16. **Future Expansion:** Smart IV Pump integration.
- 17. **Criticality Level:** Enterprise.
- 18. **Deployment Edition:** Hospital, Enterprise, AI Editions.

Module 27 – Nursing Station & Tasks

- 1. **Purpose:** The operational hub for vitals, handovers, and clinical tasks.
- 2. **Business Objectives:** Digitize the ward and ensure continuous patient care.
- 3. **Primary Actors:** Nurses, Ward In-Charge.
- 4. **Secondary Actors:** Duty Doctors.
- 5. **Key Features:** Vitals auto-flagging, task kanban board, digital shift handover.
- 6. **Major Screens:** Ward Dashboard, Vitals Entry.
- 7. **Major Reports:** Nursing Activity Report, Shift Handover Log.

8. **Major Database Entities:** PatientVitals , ClinicalTask .
9. **Dependencies:** Mod 16, 28.
10. **Used By:** Mod 26, 29.
11. **Security Considerations:** Shift sign-off authentication.
12. **Audit Requirements:** Track task completion times.
13. **Mobile Readiness:** High.
14. **API Readiness:** High.
15. **AI Readiness:** High (NEWS2 automated scoring).
16. **Future Expansion:** Bedside IoT monitor integration.
17. **Criticality Level:** Enterprise.
18. **Deployment Edition:** Hospital, Enterprise, AI Editions.

Module 28 – Bed & Occupancy Management

1. **Purpose:** Real-time tracking of bed status, transfers, and housekeeping.
2. **Business Objectives:** Maximize capacity utilization and drive accurate room billing.
3. **Primary Actors:** Admission Desk, Bed Manager, Housekeeping.
4. **Secondary Actors:** Nurses.
5. **Key Features:** Visual bed board, transfer workflows, housekeeping turnaround.
6. **Major Screens:** Bed Board, Transfer Screen.
7. **Major Reports:** Occupancy Report, Length of Stay (LOS).
8. **Major Database Entities:** BedOccupancy , BedTransfer .
9. **Dependencies:** Mod 13, 15.
10. **Used By:** Mod 27, 29.
11. **Security Considerations:** Isolation/Gender restriction enforcement.
12. **Audit Requirements:** Minute-by-minute occupancy timestamps.
13. **Mobile Readiness:** Medium (Housekeeping app).
14. **API Readiness:** High.
15. **AI Readiness:** High (Predictive forecasting).
16. **Future Expansion:** AI Bed Allocation.
17. **Criticality Level:** Enterprise.
18. **Deployment Edition:** Hospital, Enterprise, AI Editions.

Module 29 – Discharge Management

1. **Purpose:** Multi-department clearance and continuity of care planning.
2. **Business Objectives:** Prevent revenue leakage and ensure safe patient transitions.
3. **Primary Actors:** Doctors, Discharge Coordinator, Billing.
4. **Secondary Actors:** Nurses, Pharmacy.
5. **Key Features:** Clearance tracker, discharge summary generation, bed release automation.
6. **Major Screens:** Discharge Dashboard, Summary Editor.
7. **Major Reports:** Discharge Register, Readmission Analysis.
8. **Major Database Entities:** DischargeRequest , DischargeClearance .
9. **Dependencies:** Mod 18, 20, 24, 28.
10. **Used By:** Mod 30, 31.
11. **Security Considerations:** Final lock e-signatures.
12. **Audit Requirements:** Track clearance bottlenecks.
13. **Mobile Readiness:** Low.
14. **API Readiness:** High.
15. **AI Readiness:** High (Readmission risk scoring).
16. **Future Expansion:** Automated Care Pathways.
17. **Criticality Level:** Enterprise.
18. **Deployment Edition:** Hospital, Enterprise, AI Editions.

Module 30 – Patient Portal & Self Service

1. **Purpose:** The patient-facing web and mobile gateway.
2. **Business Objectives:** Improve patient satisfaction and reduce administrative workload.
3. **Primary Actors:** Patients, Family Delegates.
4. **Secondary Actors:** Customer Service.
5. **Key Features:** Report vault, appointment booking, family access delegation.
6. **Major Screens:** Patient Dashboard, Report Center.
7. **Major Reports:** Portal Adoption Report.
8. **Major Database Entities:** PatientPortalAccount , PatientPortalDocument .
9. **Dependencies:** Mod 15, 16, 24, 31.
10. **Used By:** Mod 32.
11. **Security Considerations:** Strict data ownership, OTP verification.

12. **Audit Requirements:** Track report downloads.
13. **Mobile Readiness:** High (Native app target).
14. **API Readiness:** High.
15. **AI Readiness:** High (AI Health Assistant).
16. **Future Expansion:** Wearable device integration.
17. **Criticality Level:** Advanced.
18. **Deployment Edition:** Diagnostic, Hospital, Enterprise, Telemedicine, AI Editions.

Module 31 – Appointment & Follow-Up

1. **Purpose:** Advanced scheduling, waitlists, and compliance tracking.
2. **Business Objectives:** Optimize doctor schedules and ensure clinical continuity.
3. **Primary Actors:** Appointment Desk, Call Center.
4. **Secondary Actors:** Doctors, Patients.
5. **Key Features:** Multi-channel booking, waitlist auto-promotion, follow-up planner.
6. **Major Screens:** Appointment Dashboard, Schedule Setup.
7. **Major Reports:** No-Show Report, Follow-Up Compliance.
8. **Major Database Entities:** DoctorSchedule , Appointment .
9. **Dependencies:** Mod 11, 15, 17, 30.
10. **Used By:** Mod 32.
11. **Security Considerations:** Schedule override permissions.
12. **Audit Requirements:** Track reschedules and cancellations.
13. **Mobile Readiness:** High.
14. **API Readiness:** High.
15. **AI Readiness:** High (Demand prediction).
16. **Future Expansion:** WhatsApp conversational booking.
17. **Criticality Level:** Advanced.
18. **Deployment Edition:** Clinic, Hospital, Enterprise, Telemedicine, AI Editions.

Module 32 – Telemedicine & Virtual Care

1. **Purpose:** Integrated remote care and video consultations.
2. **Business Objectives:** Expand hospital reach and generate new revenue streams.
3. **Primary Actors:** Doctors, Patients, Telemed Coordinators.
4. **Secondary Actors:** Billing.

5. **Key Features:** Digital consent, virtual waiting room, remote prescribing.
6. **Major Screens:** Virtual Waiting Room, Consult Workspace.
7. **Major Reports:** Telemedicine Utilization Report.
8. **Major Database Entities:** TelemedicineSession , TelemedicineConsent .
9. **Dependencies:** Mod 18, 19, 30, 31.
10. **Used By:** None.
11. **Security Considerations:** Encrypted session links, recording governance.
12. **Audit Requirements:** Track join/leave times.
13. **Mobile Readiness:** High.
14. **API Readiness:** High (WebRTC).
15. **AI Readiness:** High (AI consultation summaries).
16. **Future Expansion:** Cross-border consultation governance.
17. **Criticality Level:** Enterprise.
18. **Deployment Edition:** Telemedicine, Enterprise, AI Editions.

Module 40 – AI Prescription Capture

1. **Purpose:** Intelligent OCR and mapping of unstructured prescriptions.
2. **Business Objectives:** Eliminate manual data entry and prevent revenue leakage.
3. **Primary Actors:** Reception, Billing.
4. **Secondary Actors:** AI Verification Officer.
5. **Key Features:** OCR extraction, smart service mapping, duplicate detection.
6. **Major Screens:** Upload Screen, Verification Workbench.
7. **Major Reports:** AI Accuracy Report, Revenue Impact.
8. **Major Database Entities:** AIPrescriptionCapture , AIInvestigationMapping .
9. **Dependencies:** Mod 10, 15, 20.
10. **Used By:** Mod 21, 25.
11. **Security Considerations:** PHI protection in image storage.
12. **Audit Requirements:** Track human overrides of AI predictions.
13. **Mobile Readiness:** Medium (Mobile upload).
14. **API Readiness:** High.
15. **AI Readiness:** High (Core AI Module).
16. **Future Expansion:** LLM-assisted clinical interpretation.
17. **Criticality Level:** AI.

18. Deployment Edition: AI Edition.

SECTION 4: MODULE GROUP SUMMARIES

- **Foundation Modules (01-06):** The bedrock of the SaaS platform. These modules handle multi-tenancy, security, auditing, and localization. They operate invisibly behind the scenes to keep the system secure and scalable.
 - **Master Data Modules (07-14):** The dictionaries of the hospital. They define the physical locations, clinical departments, billable services, and inventory items required before any patient transaction can occur.
 - **Patient Journey Modules (15-17, 31):** Manage the patient's entry into the system, ensuring unique identification (MPI) and orchestrating their physical and digital appointments.
 - **Clinical Modules (18-20):** The physician's digital workspace. These modules replace paper charts with structured encounters, diagnoses, and electronic prescriptions.
 - **Diagnostic Modules (21-24):** The Laboratory Information System (LIS). A strict pipeline from barcode generation to pathologist verification and secure report delivery.
 - **Radiology Modules (25):** The Radiology Information System (RIS). Manages expensive imaging assets, technician workflows, and structured radiologist dictations.
 - **IPD Modules (26-29):** The logistical and clinical management of admitted patients. Covers bed occupancy, nursing tasks, medication administration, and the complex discharge clearance process.
 - **Finance Modules:** (*Integrated throughout*). Modules 10, 28, 29, and 40 contain heavy financial touchpoints for billing, pricing, and revenue assurance.
 - **Patient Engagement Modules (30):** The digital front door, empowering patients with self-service access to their health data and hospital services.
 - **Telemedicine Modules (32):** Extends clinical care beyond the hospital walls via secure video consultations and remote prescribing.
 - **AI Modules (40):** The intelligence layer that automates administrative burdens, starting with the conversion of unstructured prescription images into structured billing orders.
-

SECTION 5: CROSS MODULE WORKFLOWS

Standard OPD to Lab Workflow

```

[15] Patient Registration
    |
    v
[31] Appointment Booking
    |
    v
[18] Doctor Encounter (Orders Lab Tests)
    |
    v
[21] Sample Collection (Generates Barcode)
    |
    v
[22] Result Entry (Machine/Manual)
    |
    v
[23] Result Verification (Pathologist)
    |
    v
[24] Report Release
    |
    v
[30] Patient Portal (Patient views PDF)
  
```

Telemedicine Workflow

```

[30] Patient Portal (Books Telemed)
    |
    v
[31] Appointment (Schedules Slot)
    |
    v
[32] Telemedicine (Digital Consent & Waiting Room)
    |
    v
[18] Encounter (Doctor documents via Video Call)
    |
    v
  
```

```
[19] Prescription (ePrescription generated)
    |
    v
[30] Patient Portal (Patient downloads Rx)
```

AI Prescription Flow

```
[30] Patient Portal (Uploads External Rx Image)
    |
    v
[40] AI Prescription Capture (OCR & Extraction)
    |
    v
[40] Smart Mapping (Maps to [10] Service Catalog)
    |
    v
[40] Verification Workbench (Human approves)
    |
    v
[21] Sample Collection (Orders automatically created)
```

SECTION 6: MODULE DEPENDENCY HEATMAP

Module Group	Dependency Level	Description
Foundation (01-06)	Independent	Relies on nothing. Everything relies on them.
Master Data (07-14)	Low Dependency	Relies only on Foundation.
Patient Journey (15-17)	Medium Dependency	Relies on Foundation and Master Data.
Clinical (18-20)	High Dependency	Relies on Patient Journey and Master Data.
Diagnostics (21-25)	High Dependency	Strict linear dependency within the group.
IPD (26-29)	Very High Dependency	Relies on Clinical, Master Data, and Patient.
Engagement/Telemed (30-32)	High Dependency	Sits on top of all clinical and diagnostic data.
AI (40)	Very High Dependency	Requires mature Master Data to map against.

SECTION 7: EDITION MAPPING

Module	Clinic	Diagnostic	Hospital	Enterprise	Telemed	AI Ed.
01-06 (Foundation)	✓	✓	✓	✓	✓	✓
07-12 (Masters)	✓	✓	✓	✓	✓	✓
13 (Ward Setup)			✓	✓		✓
14 (Diag Inventory)		✓	✓	✓		✓
15-17 (Patient/Appt)	✓	✓	✓	✓	✓	✓
18-20 (Clinical)	✓		✓	✓	✓	✓
21-24 (Laboratory)		✓	✓	✓		✓
25 (Radiology)		✓	✓	✓		✓
26-29 (IPD)			✓	✓		✓
30 (Patient Portal)		✓	✓	✓	✓	✓
31 (Adv. Appt)	✓		✓	✓	✓	✓
32 (Telemedicine)				✓	✓	✓
40 (AI Prescription)						✓

SECTION 8: IMPLEMENTATION PRIORITY

Based on the official Development Sequence Guide:

- **Phase 1 (The Bedrock):** Modules 01, 02, 03, 04, 06.

- **Phase 2 (The Dictionaries):** Modules 07, 08, 09, 10, 11, 12, 13, 14, 20.
 - **Phase 3 (The Front Desk):** Modules 15, 16, 17, 05.
 - **Phase 4 (Clinical & Diagnostics):** Modules 18, 19, 21, 22, 23, 24, 25.
 - **Phase 5 (The Ward & Edge):** Modules 28, 27, 26, 29, 30, 31, 32, 40.
-

SECTION 9: PRODUCT MATURITY MAP

- **Foundation:** Modules 01-14. The mandatory base.
 - **Core:** Modules 15-21. The minimum viable healthcare operations.
 - **Advanced:** Modules 22-25, 30, 31. Automation and patient-facing tools.
 - **Enterprise:** Modules 26-29, 32. Complex inpatient logistics and virtual care.
 - **AI:** Module 40. Next-generation machine learning automation.
-

SECTION 10: CONCLUSION

The ABSHealthcareLite Functional Module Catalog demonstrates a platform of immense **breadth** (covering everything from basic OPD to complex IPD and Telemedicine) and profound **depth** (featuring enterprise-grade auditability, multi-level approvals, and strict clinical safety gates).

Our **Competitive Strengths** lie in our unified architecture. Unlike fragmented hospital systems that bolt a third-party LIS onto an EMR, ABSHealthcareLite natively integrates diagnostics, pharmacy, and patient engagement. The multi-tenant SaaS design ensures rapid scalability, while the localization engine guarantees global market readiness.

The **Future Roadmap** is secured by an architecture that preserves current stability (SQL Server/WebForms) while explicitly paving the way for modern API-driven, mobile-first, and AI-augmented healthcare delivery.

ABSHealthcareLite Product Master Book

Volume 3 – Implementation, Deployment & Go-Live Guide

SECTION 1: EXECUTIVE SUMMARY

The **ABSHealthcareLite Product Master Book Volume 3 – Implementation, Deployment & Go-Live Guide** is the definitive operational manual for transitioning a healthcare facility from legacy systems (or paper) to the ABSHealthcareLite platform.

While **Volume 1** outlines the product vision and **Volume 2** details the functional capabilities, **Volume 3** provides the tactical, step-by-step roadmap required to successfully deploy the software. Designed for Hospital Owners, Project Managers, and SaaS Operations Teams, this guide ensures that implementations are structured, secure, clinically safe, and minimally disruptive to daily hospital operations.

This guide strictly adheres to the platform's core architectural tenets: Multi-tenant SaaS design, robust localization (RTL/LTR), and a future-proof technology stack (SQL Server to PostgreSQL, WebForms to MVC/Core APIs).

SECTION 2: IMPLEMENTATION METHODOLOGY

A successful ERP rollout requires a disciplined, phased approach to mitigate clinical and operational risks.

- **Phase 1: Planning:** Define scope, establish the project team, map existing workflows to ABSHealthcareLite modules, and set the Go-Live date.
- **Phase 2: System Configuration:** Provision the tenant, configure global settings, localization, and establish the physical location hierarchy (Branches/Wards).
- **Phase 3: Master Data Setup:** Populate the critical dictionaries (Services, Doctors, Pharmacy Catalog, Diagnostic Inventory).
- **Phase 4: User Training:** Role-specific, hands-on training for all hospital staff using a sandbox environment.

- **Phase 5: Pilot Operation:** Run a controlled subset of operations (e.g., one specific OPD clinic) through the system to validate workflows.
- **Phase 6: Go Live:** The official cutover. Legacy systems become read-only; ABSHealthcareLite becomes the system of record.
- **Phase 7: Hypercare Support:** Intensive, on-site IT support for the first 2-4 weeks post-Go-Live to resolve immediate user friction.
- **Phase 8: Continuous Improvement:** Post-stabilization phase to activate advanced modules (e.g., Patient Portal, Telemedicine, AI Prescription Capture).

SECTION 3: DEPLOYMENT MODELS

ABSHealthcareLite supports flexible hosting architectures depending on client size and regulatory requirements:

- **Cloud SaaS:** The default model. Multi-tenant architecture hosted on public cloud (AWS/Azure). Fastest deployment, lowest upfront cost, managed by ABS.
- **Private Cloud:** A dedicated, single-tenant cloud instance for large enterprise networks requiring strict data sovereignty.
- **On-Premise:** Deployed on local hospital servers. Recommended only for regions with highly unstable internet or strict local data residency laws.
- **Hybrid:** Core transactional database on-premise, with cloud-bursting for the Patient Portal and Telemedicine modules.
- **Multi-Branch Deployment:** A single tenant (`CompanyId`) managing multiple physical hospitals (`BranchId`), sharing a unified Master Patient Index (MPI).
- **Multi-Hospital Group Deployment:** A corporate entity managing multiple distinct hospital brands as separate `CompanyId` tenants under a single SaaS instance.

SECTION 4: INFRASTRUCTURE REQUIREMENTS

(Specifications for On-Premise or Private Cloud deployments. SaaS clients require only standard broadband and modern web browsers).

- **Minimum:**
 - App Server: 4 Core CPU, 16GB RAM.
 - DB Server: 8 Core CPU, 32GB RAM (SQL Server Standard).

- **Recommended:**
 - App Server: 8 Core CPU, 32GB RAM.
 - DB Server: 16 Core CPU, 64GB RAM (SQL Server Enterprise).
 - **Enterprise:** Load-balanced App Server cluster, Clustered DB with Always-On Availability Groups, dedicated Reporting Server.
 - **Storage:** SSD/NVMe for active databases; tiered cloud storage for heavy DICOM/PDF archival.
 - **Network:** Dedicated leased line (100Mbps+) with 4G/5G failover.
 - **Security:** TLS 1.2/1.3 SSL certificates, Web Application Firewall (WAF).
 - **Future Readiness:** Infrastructure must support reverse proxies and containerization (Docker/Kubernetes) for the upcoming MVC/Core/API migration.
-

SECTION 5: INITIAL SYSTEM SETUP

The foundational steps to initialize a new hospital environment:

1. **System Installation:** Deploy application binaries and execute database creation scripts.
 2. **Company Creation:** Generate the unique `CompanyId` (Tenant).
 3. **License Setup:** Activate purchased modules (e.g., Hospital Edition vs. Diagnostic Edition).
 4. **Branch Setup:** Define the physical locations (Module 07).
 5. **Department Setup:** Create clinical and operational departments (Module 08).
 6. **Basic Configuration:** Set timezone, default currency, and primary language (Module 06).
-

SECTION 6: SECURITY IMPLEMENTATION

Protecting Patient Health Information (PHI) is paramount before any real data is entered.

1. **Users:** Create staff accounts (Module 02).
2. **Roles:** Define job-specific roles (e.g., "Junior Nurse", "Senior Pathologist").
3. **Permissions:** Map granular access rights to roles (Module 03).
4. **Password Policies:** Enforce complexity, expiration, and lockout rules.
5. **MFA Readiness:** Configure Multi-Factor Authentication for remote access.

6. **Audit Policies:** Ensure Module 04 (Audit Center) is actively logging all CRUD operations.
 7. **Data Access Policies:** Enforce branch-level and department-level visibility restrictions.
-

SECTION 7: MASTER DATA IMPLEMENTATION

The system cannot function without its dictionaries. This is often the most time-consuming phase.

1. **Departments & Categories:** Establish the reporting hierarchy.
 2. **Services:** Populate the Master Service Catalog (Module 10) with exact pricing.
 3. **Doctors:** Create provider profiles and schedules (Module 11).
 4. **Referrals:** Import external referring physician networks (Module 12).
 5. **Sample Types:** Define lab containers and routing rules.
 6. **Investigation Catalog:** Map lab tests and radiology procedures.
 7. **Medication Catalog:** Import the pharmacy formulary (Module 20).
-

SECTION 8: PATIENT DATA MIGRATION

Transitioning historical data from legacy systems.

- **Migration Strategy:** Extract, Transform, Load (ETL) via standardized CSV/Excel templates or direct API scripts.
 - **Patient Master:** Import demographic data to establish the MPI.
 - **Doctors:** Import historical provider lists to maintain record integrity.
 - **Historical Reports:** Migrate past PDF lab/rad reports into the Health Document Vault.
 - **Historical Bills:** Migrate outstanding balances into the Patient Ledger.
 - **Validation Rules:** Run scripts to detect duplicate patients or missing mandatory fields.
 - **Migration Audit:** Generate a sign-off report comparing legacy record counts to imported ABSHealthcareLite counts.
-

SECTION 9: MODULE IMPLEMENTATION SEQUENCE

Rollout should follow the **Development Sequence Guide (Volume 1, Doc 02)** to prevent workflow blockers:

1. **Phase 1:** Foundation & Security (01-06).
 2. **Phase 2:** Master Data (07-14, 20).
 3. **Phase 3:** Patient Identity & Basic Scheduling (15-17).
 4. **Phase 4:** Clinical OPD & Prescriptions (18-19).
 5. **Phase 5:** Laboratory & Radiology (21-25).
 6. **Phase 6:** IPD, Nursing, MAR, & Discharge (26-29).
 7. **Phase 7:** Patient Portal & Telemedicine (30-32).
 8. **Phase 8:** AI Prescription Capture (40).
-

SECTION 10: TRAINING FRAMEWORK

Training must be role-specific and conducted in a safe "Sandbox" environment.

- **Reception Training:** Patient registration, appointment booking, queue management.
 - **Doctor Training:** EMR navigation, clinical notes, ePrescribing, result review.
 - **Lab Training:** Barcode scanning, result entry, critical value escalation.
 - **Radiology Training:** Modality scheduling, structured dictation.
 - **Nursing Training:** Ward dashboard, vitals entry, MAR execution, shift handover.
 - **Billing Training:** Charge capture, discounts, discharge settlement.
 - **Admin/Management Training:** User management, master data updates, dashboard analytics.
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SECTION 11: TESTING FRAMEWORK

- **Unit Testing:** Developer-led testing of individual functions.
- **Integration Testing:** Ensuring data flows correctly (e.g., Doctor orders Lab → Lab receives order → Billing captures charge).
- **UAT (User Acceptance Testing):** Hospital staff execute real-world scenarios in the sandbox and sign off on functionality.

- **Performance Testing:** Simulating peak load (e.g., 8:00 AM OPD rush) to ensure server stability.
 - **Security Testing:** Vulnerability scanning and RBAC verification.
 - **Disaster Recovery Testing:** Simulating a server failure to verify backup restoration times.
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SECTION 12: GO-LIVE READINESS CHECKLIST

A strict gatekeeping process before flipping the switch.

- **Technical Checklist:** Servers provisioned, SSL active, backups configured, printers/barcode scanners tested.
 - **Operational Checklist:** Master data 100% loaded, user accounts created, legacy system set to read-only.
 - **Clinical Checklist:** Doctors trained, prescription templates verified, lab reference ranges validated.
 - **Billing Checklist:** Service pricing verified, tax rules configured.
 - **Security Checklist:** Default passwords changed, RBAC audited.
 - **Management Approval Checklist:** Formal sign-off from Hospital CEO and IT Director.
-

SECTION 13: GO-LIVE EXECUTION PLAN

- **Day -30:** Freeze master data changes in legacy system. Begin UAT.
 - **Day -15:** Complete user training. Finalize infrastructure.
 - **Day -7:** Execute final data migration (Delta load).
 - **Day -1:** System downtime window. Final technical checks.
 - **Go Live Day:** ABSHealthcareLite is active. Hypercare team on the floor.
 - **Day +7:** First major billing reconciliation. Resolve immediate workflow bottlenecks.
 - **Day +30:** Transition from Hypercare to standard SLA support.
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SECTION 14: POST GO-LIVE SUPPORT

- **Hypercare:** Dedicated, on-site (or dedicated remote) team for the first 2-4 weeks to provide immediate shoulder-to-shoulder support.
 - **Issue Tracking:** Centralized ticketing system for logging bugs and feature requests.
 - **Priority Levels:** P1 (System Down), P2 (Critical Workflow Blocked), P3 (Minor Bug), P4 (Cosmetic/Enhancement).
 - **Escalation Matrix:** Clear pathways from L1 Helpdesk to L3 Engineering.
 - **Knowledge Base:** Access to digital manuals, video tutorials, and FAQs.
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SECTION 15: BACKUP & DISASTER RECOVERY

- **Backup Policy:** Automated daily full backups, hourly differential backups, and 15-minute transaction log backups.
 - **Retention Policy:** Keep daily backups for 30 days, monthly backups for 1 year, yearly backups for 7 years (or per local law).
 - **Recovery Testing:** Mandatory quarterly restoration drills to verify data integrity.
 - **Business Continuity:** For on-premise, maintain a hot-standby server. For SaaS, utilize multi-zone cloud replication.
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SECTION 16: MULTI-TENANT SAAS OPERATIONS

For the ABS Operations Team managing the cloud infrastructure:

- **Tenant Provisioning:** Automated scripts to spin up a new `CompanyId` and default configurations within minutes.
- **Tenant Configuration:** Managing feature flags (e.g., enabling Telemedicine for Tenant A, disabling for Tenant B).
- **Tenant Monitoring:** Centralized APM (Application Performance Monitoring) to detect slow queries or high resource usage per tenant.
- **Tenant Upgrade Strategy:** Zero-downtime deployments utilizing database schema versioning.
- **Tenant Data Isolation:** Automated daily audits to ensure no cross-tenant data leakage in SQL queries.

SECTION 7: LOCALIZATION IMPLEMENTATION

- **Languages:** English, Bangla, Arabic, Urdu, Hindi.
 - **RTL Testing:** Rigorous UI testing to ensure Arabic/Urdu selections correctly mirror the layout (menus, grids, modals) right-to-left.
 - **LTR Testing:** Standard left-to-right verification.
 - **Resource Files:** Ensuring all custom hospital data (e.g., Ward Names) are entered in the primary operating language.
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SECTION 18: MOBILE & API READINESS

- **Patient App:** Deploying the patient portal as a PWA or Native App.
 - **Doctor App:** Enabling mobile EMR access for ward rounds.
 - **Future Nursing App:** Preparing the network infrastructure (Wi-Fi coverage) for bedside tablet deployment.
 - **REST APIs:** Exposing secure endpoints for third-party integrations (e.g., accounting software).
 - **FHIR Readiness:** Ensuring data mappings align with Fast Healthcare Interoperability Resources standards for future health exchange integration.
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SECTION 19: AI IMPLEMENTATION STRATEGY

- **Module 40 Rollout:** Do not activate AI Prescription Capture until the Master Service Catalog and Pharmacy Catalog have been stable for at least 30 days.
 - **AI Governance:** Establish the AI Review Committee to monitor accuracy.
 - **Human Verification:** Train the AI Verification Officer on the mandatory human-in-the-loop workflow.
 - **Model Monitoring:** Track OCR accuracy and extraction confidence scores weekly.
 - **Clinical Safety:** Ensure critical risk items (e.g., high-alert meds) always force manual review.
-

SECTION 20: HOSPITAL GO-LIVE MATURITY MODEL

Hospitals evolve in their use of ABSHealthcareLite. Implementations should target a specific level and grow over time:

- **Level 1 (Basic Clinic):** OPD Registration, Appointments, Basic Billing.
- **Level 2 (Diagnostic Center):** Level 1 + LIS/RIS, Barcodes, Report Release.
- **Level 3 (General Hospital):** Level 2 + IPD, Bed Occupancy, Nursing Station, MAR.
- **Level 4 (Multi-Branch Hospital):** Level 3 + Multi-location routing, centralized procurement.
- **Level 5 (Enterprise Healthcare Network):** Level 4 + Patient Portal, Telemedicine, AI Automation, Advanced Analytics.

SECTION 21: IMPLEMENTATION RISKS & MITIGATION

- **Technical Risks:** Legacy data corruption. *Mitigation: Rigorous ETL validation scripts.*
- **Operational Risks:** Front desk bottlenecks during Go-Live. *Mitigation: Extra staffing and Hypercare support during week 1.*
- **Clinical Risks:** Doctors refusing to use ePrescriptions. *Mitigation: Strong executive sponsorship, easy-to-use templates, and AI Prescription Capture (Module 40) as a fallback.*
- **Financial Risks:** Billing errors due to incorrect Master Catalog pricing. *Mitigation: Mandatory Finance Director sign-off on the pricing matrix during UAT.*
- **Change Management Risks:** Staff resistance to new workflows. *Mitigation: Early inclusion of key users in the planning phase; comprehensive training.*

SECTION 22: FINAL IMPLEMENTATION ROADMAP

```
=====
ABSHealthcareLite Implementation Roadmap
=====
```

```
[ PHASE 1: PLANNING ]
```

```
Project Kickoff -> Workflow Mapping -> Infrastructure Audit -> Go-Live Date Se
```

```

      |
      v
[ PHASE 2: CONFIGURATION ]
  Tenant Provisioning -> Localization Setup -> Branch/Ward Hierarchy Built
      |
      v
[ PHASE 3: DATA MIGRATION & MASTERS ]
  Service Catalog Load -> Pharmacy Catalog Load -> Legacy Patient Data Import
      |
      v
[ PHASE 4: TRAINING & UAT ]
  Role-Based Sandbox Training -> User Acceptance Testing -> Executive Sign-Off
      |
      v
[ PHASE 5: PILOT & PRE-LIVE ]
  Single Clinic Pilot -> Final Data Sync -> Legacy System Freeze
      |
      v
[ PHASE 6: GO-LIVE ]
  System Active -> Hypercare Team Deployed -> Real-Time Issue Resolution
      |
      v
[ PHASE 7: OPTIMIZATION & AI ]
  Hypercare Ends -> Portal/Telemed Activated -> AI Module 40 Enabled -> SLA Support

=====

```

SECTION 23: APPENDIX A – IMPLEMENTATION CHECKLISTS

- **IT Infrastructure Checklist:** Servers, Network, Printers, Scanners, SSL.
 - **Master Data Checklist:** Departments, Services, Prices, Doctors, Drugs, Wards, Beds.
 - **Security Checklist:** Users, Roles, Passwords, Branch Access.
 - **Training Checklist:** Attendance logs, Competency tests passed.
-

SECTION 24: APPENDIX B – GO-LIVE SIGN-OFF TEMPLATES

Formal documents requiring physical or digital signatures to proceed to Go-Live:

- **IT Sign-Off:** "Infrastructure, backups, and security configurations are verified."
 - **Clinical Sign-Off:** "Prescription templates, lab reference ranges, and MAR workflows are verified safe for patient care."
 - **Billing Sign-Off:** "Service pricing, tax rules, and discount matrices are accurate."
 - **Management Sign-Off:** "The facility is operationally ready to transition to ABSHealthcareLite."
-

SECTION 25: CONCLUSION

The implementation of ABSHealthcareLite is a transformative event for a healthcare facility. By adhering to this structured guide, implementation teams can ensure a transition that prioritizes **clinical safety**, achieves **operational excellence**, and establishes a foundation for **future scalability**.

Whether deploying a standalone diagnostic center or a multi-branch enterprise hospital, this methodology guarantees that the platform's multi-tenant architecture, multilingual capabilities, and advanced AI features are leveraged to their maximum potential, ultimately driving better patient outcomes and sustainable business growth.

ABSHealthcareLite Product Master Book

Volume 4 – UI/UX Blueprint & Design System

SECTION 1: EXECUTIVE SUMMARY

The ABSHealthcareLite Product Master Book Volume 4 – UI/UX Blueprint & Design System serves as the definitive guide for the user experience (UX), user interface (UI), and interaction design of the ABSHealthcareLite platform.

- **Purpose:** To establish a unified, scalable, and clinically safe design language that ensures consistency across all modules, tenants, and devices.
 - **Relationship with Product Books:** While Volume 1 outlines the vision, Volume 2 details the functional catalog, and Volume 3 provides the implementation roadmap, Volume 4 dictates *how the user interacts* with those functions.
 - **Relationship with Architecture Documents:** This blueprint aligns with the System Layer Architecture, ensuring that UI components are decoupled from business logic, supporting the transition from ASP.NET WebForms to future MVC/Core and native mobile platforms, while strictly adhering to Multi-Tenant SaaS and Multilingual (RTL/LTR) requirements.
-

SECTION 2: DESIGN PHILOSOPHY

The ABSHealthcareLite design philosophy is built on ten core principles:

1. **Healthcare First:** Clinical safety supersedes aesthetics. Critical patient data (allergies, alerts) must never be hidden behind clicks.
2. **Speed First:** High-volume areas (Registration, Billing, Pharmacy) must support rapid data entry.
3. **Safety First:** Destructive actions (which are soft-deletes) and critical clinical decisions require explicit confirmation.
4. **Low Click Operations:** Optimize workflows to minimize mouse travel and click fatigue.

5. **Keyboard Friendly:** Power users must be able to navigate, enter data, and save forms using only the keyboard.
 6. **Mobile Ready:** All patient-facing and physician-facing interfaces must be responsive by default.
 7. **Accessibility Ready:** High contrast, readable typography, and clear visual hierarchy.
 8. **Multi-Language Ready:** UI must gracefully handle text expansion and contraction across languages.
 9. **RTL Ready:** Native support for Right-to-Left mirroring for Arabic and Urdu.
 10. **AI Ready:** AI suggestions must be clearly distinguishable from human-entered data, requiring explicit human verification.
-

SECTION 3: GLOBAL DESIGN SYSTEM

Color System

- **Primary Brand Color:** Customizable per Tenant (Company Branding Engine). Default: Medical Blue.
- **Secondary Color:** Neutral Gray for structural elements.
- **Status Colors:**
 - *Success/Active:* Green
 - *Warning/Pending:* Amber/Yellow
 - *Danger/Critical:* Red
 - *Info/Draft:* Light Blue

Typography

- **Font Family:** System-native sans-serif (e.g., Roboto, Segoe UI, San Francisco) to ensure fast loading and optimal multilingual rendering (e.g., Noto Sans Arabic, Noto Sans Bengali).
- **Hierarchy:** Clear distinction between H1 (Page Titles), H2 (Section Headers), Body Text, and Micro-copy.

Spacing & Layout

- 8px/16px/24px grid system.
- High-density mode for clinical/billing grids; comfortable mode for patient portals.

Icons & Badges

- Universal, medically recognized iconography (e.g., standard allergy, biohazard, and prescription symbols).
- Badges used for counts (e.g., Unread Notifications, Pending Approvals).

Alerts & Forms

- Toast notifications for transient success messages.
- Persistent banners for critical patient alerts (e.g., "HIGH FALL RISK").
- Forms use top-aligned labels for faster scanning and better RTL translation.

SECTION 4: MULTI-LANGUAGE UI STANDARDS

ABSHealthcareLite is a global platform supporting:

- **English** (LTR)
- **Bangla** (LTR)
- **Arabic** (RTL)
- **Urdu** (RTL)
- **Hindi** (LTR)

RTL Rules:

- When Arabic or Urdu is selected, the entire UI matrix mirrors. Menus move to the right, text aligns right, and back/forward arrows reverse direction.
- Numeric inputs (e.g., phone numbers, lab results) remain LTR even in RTL layouts.

Text Expansion Handling:

- UI containers must allow for up to 30% text expansion, as translated words (e.g., English to Bangla) often require more horizontal space.
- No hardcoded fixed widths for buttons or labels.

Language Switcher:

- Available globally in the top navigation bar, applying changes instantly via resource files without requiring a hard logout.

SECTION 5: RESPONSIVE DESIGN STANDARDS

- **Desktop (1920x1080+):** Optimized for complex grids, multi-panel clinical views, and high-density data entry.
 - **Tablet (768x1024 to 1024x1366):** Optimized for touch. Used heavily in Nursing Stations, Bedside MAR, and Doctor rounds.
 - **Mobile (375x667 to 428x926):** Optimized for single-column vertical scrolling. Target for Patient Portal, Telemedicine, and Doctor quick-approvals.
 - **Kiosk (1080x1920 Portrait):** High-contrast, large touch targets for self-check-in and queue token generation.
-

SECTION 6: NAVIGATION STANDARDS

- **Host Admin Navigation:** Global system settings, tenant provisioning, cross-tenant analytics. (Left Sidebar).
 - **Tenant Admin Navigation:** Hospital-specific master data, user management, financial settings. (Left Sidebar).
 - **Doctor Navigation:** Patient-centric. Worklist, EMR, ePrescription, Telemedicine. (Top-level tabs with left-side patient context).
 - **Nurse Navigation:** Ward-centric. Bed board, Task list, MAR. (Visual grid navigation).
 - **Reception Navigation:** Fast-action centric. Registration, Appointment, Billing. (Keyboard-shortcut optimized).
 - **Lab/Radiology Navigation:** Queue-centric. Pending samples, Result entry, Verification.
 - **Patient Portal Navigation:** Simple, consumer-friendly bottom tab bar (Mobile) or top header (Desktop).
-

SECTION 7: STANDARD SCREEN TYPES

1. **List Screen:** (e.g., Patient List). *Layout:* Search/Filter top, Data Grid middle, Pagination bottom. *Actions:* Add New, Export, Bulk Action.
2. **Entry Screen:** (e.g., Add User). *Layout:* Form fields grouped by logical sections. *Actions:* Save, Save & New, Cancel.

3. **Dashboard Screen:** (e.g., Executive KPI). *Layout:* Widget/Card based. *Actions:* Date range filter, Drill-down.
4. **Approval Screen:** (e.g., Result Verification). *Layout:* Split screen (Data on left, Approval controls on right). *Actions:* Approve, Reject, Return for Correction.
5. **Wizard Screen:** (e.g., Initial Tenant Setup). *Layout:* Progress tracker top, Step content middle, Prev/Next bottom.
6. **Report Screen:** (e.g., Daily Collection). *Layout:* Parameter selection left/top, Report viewer right/bottom.
7. **Print Screen:** Clean, unstyled layout optimized for A4/A5 physical printing with header/footer branding.
8. **Portal/Mobile Screen:** Consumer-grade, large touch targets, simplified terminology.

SECTION 8: FORM DESIGN STANDARDS

- **Patient Forms:** Must clearly separate Demographics, Clinical History, and Billing Info.
- **Billing Forms:** Must feature a persistent "Total/Due" sticky footer.
- **Prescription Forms:** Generic-first search, structured dosage dropdowns, prominent allergy warnings.
- **Lab Forms:** Fast numeric keypad entry for results, visual flags for out-of-range values.
- **Admission Forms:** Bed selection visual map, deposit collection integration.
- **Validation Rules:** Inline real-time validation (red borders for errors) before form submission. Mandatory fields marked with an asterisk (*).

SECTION 9: DATA GRID STANDARDS

- **Search:** Global quick search per grid.
- **Filter:** Advanced collapsible filter panels for complex queries.
- **Sort:** Clickable column headers with visual ascending/descending indicators.
- **Pagination:** Standardized page sizes (10, 25, 50, 100) with total record counts.
- **Export:** Universal export to Excel, CSV, and PDF.
- **Bulk Actions:** Checkbox column for multi-select (e.g., Bulk Approve).
- **Column Personalization:** Users can hide/show/reorder columns, saved to their User Profile preferences.

SECTION 10: DASHBOARD DESIGN STANDARDS

- **Executive Dashboard:** High-level KPIs (Revenue, Occupancy, Patient Volume). Visual charts (Pie, Bar, Line).
 - **Doctor Dashboard:** Clinical focus. Today's appointments, pending reports, critical alerts.
 - **Lab Dashboard:** Operational focus. TAT (Turnaround Time) gauges, pending verifications, machine status.
 - **Billing Dashboard:** Financial focus. Daily collection, outstanding dues, discount summaries.
 - **Patient Dashboard:** Health focus. Upcoming appointments, recent lab results, active medications.
 - **Host Dashboard:** SaaS focus. Active tenants, system health, license expirations.
-

SECTION 11: WORKFLOW UX STANDARDS

Optimal click paths designed for speed:

- **Patient Registration:** Mobile No. -> Check MPI -> Auto-fill -> Save. (Target: < 30 seconds).
 - **Appointment:** Select Doctor -> Select Date -> Click Slot -> Confirm. (Target: 4 clicks).
 - **Prescription:** Select Patient -> Add Dx -> Search Generic -> Select Route/Dose/Duration -> Save & Print.
 - **Investigation Billing:** Scan Barcode/Enter PID -> Select Tests -> Collect Payment -> Print Receipt & Barcode.
 - **Admission:** ER/OPD Request -> Select Bed -> Collect Deposit -> Generate IPD No.
 - **Discharge:** Clinical Clearance -> Pharmacy Clearance -> Final Bill Generation -> Settlement -> Release Bed.
 - **Telemedicine:** Click Link -> Device Test -> Virtual Waiting Room -> Join Call.
 - **AI Verification:** AI Extracts Data -> Highlights mapped fields -> User reviews side-by-side -> Clicks "Verify & Save".
-

SECTION 12: ACCESSIBILITY STANDARDS

- **Keyboard Navigation:** Full `Tab` index support. `Enter` to submit, `Esc` to close modals.
 - **Color Contrast:** Minimum WCAG AA compliance (4.5:1 ratio) for text against backgrounds.
 - **Screen Reader Readiness:** ARIA labels on all critical functional icons and dynamic alerts.
 - **Touch Target Sizes:** Minimum 44x44 pixels for all actionable elements on mobile/tablet views.
 - **Accessibility Compliance Readiness:** Architecture designed to support future strict government accessibility mandates.
-

SECTION 13: AUDIT & SECURITY UX

- **Approval Indicators:** Digital signature badges showing "Verified by Dr. X on [Date]".
 - **Permission Indicators:** Disabled/Grayed-out buttons with tooltip "Insufficient Permission" rather than hiding the button entirely, to aid discoverability.
 - **Audit Indicators:** A standard "History" or "Log" icon on every major entity to view the CRUD audit trail.
 - **Sensitive Data Indicators:** PII (Personally Identifiable Information) masking (e.g., `+1-***-***-7890`) with a "Click to Reveal" eye icon for authorized users.
 - **AI Decision Indicators:** AI-generated suggestions must have a distinct visual treatment (e.g., a sparkle icon or purple border) and a confidence score percentage.
-

SECTION 14: NOTIFICATION UX

- **Alert Priorities:**
 - *Critical (Red):* Panic values, system downtime. Modal popup requiring acknowledgment.
 - *High (Orange):* Bed transfer requests, STAT orders. Persistent top banner.
 - *Info (Blue):* Normal lab result ready, appointment booked. Toast notification.
 - **Channels:** UI integrates seamlessly with SMS, Email, WhatsApp, Portal, and Push Notification dispatchers, showing delivery status logs.
-

SECTION 15: MODULE SCREEN CATALOG

Summary of core screens per module (01-32, 40):

- **01 Company/Tenant:** Tenant List, Tenant Setup Wizard, Branding Configuration.
- **02 User Management:** User Grid, User Profile Entry, Password Reset Modal.
- **03 Role Permission:** Role List, Permission Matrix Grid.
- **04 Audit Center:** Global Audit Log Grid, Entity History Modal.
- **05 Notification Center:** Notification Template Builder, Dispatch Log.
- **06 Localization:** Dictionary Grid, Language Switcher UI.
- **07 Branch Location:** Branch List, Branch Details.
- **08 Department:** Department Hierarchy Tree, Department Entry.
- **09 Category:** Category List, Sub-category mapping.
- **10 Master Service:** Service Catalog Grid, Pricing Matrix, CPT Mapping.
- **11 Doctor:** Doctor Profile, Schedule Builder, Commission Setup.
- **12 Referral Doctor:** Referral List, Referral Portal Dashboard.
- **13 Ward Cabin Bed:** Visual Bed Board, Ward Entry.
- **14 Diagnostic Inventory:** Stock Ledger, PO Wizard, GRN Entry.
- **15 Patient Registration:** MPI Search, Registration Form, Smart Card Print.
- **16 Patient Profile Ledger:** Patient 360 Dashboard, Ledger Grid.
- **17 Appointment:** Calendar View, Queue Display Board, Token Print.
- **18 Doctor Worklist:** OPD Queue List, Clinical Encounter Workspace.
- **19 Prescription:** CPOE Interface, Drug Interaction Alert Modal, Rx Print.
- **20 Pharmacy:** POS Screen, Dispense Queue, Batch Expiry Dashboard.
- **21 Sample Collection:** Phlebotomy Worklist, Barcode Generation, Acknowledgement Print.
- **22 Result Entry:** Analyzer Interface, Manual Entry Grid, Delta Check Alerts.
- **23 Result Verification:** Pathologist Worklist, Side-by-Side Verification Screen.
- **24 Report Release:** Delivery Dashboard, PDF Viewer, Email/WhatsApp Dispatch.
- **25 Radiology:** Modality Worklist, DICOM Viewer Integration, Dictation Interface.
- **26 MAR:** Bedside Tablet View, Med Administration Grid, Witness Sign-off Modal.
- **27 Nursing Station:** Ward Dashboard, Vitals Flowsheet, Task List.
- **28 Bed Occupancy:** Real-time Occupancy Heatmap, Transfer Wizard.

- **29 Discharge:** Discharge Summary Builder, Clearance Checklist, Final Bill.
- **30 Patient Portal:** Mobile-first Home, Appointment Booker, Report Downloader.
- **31 Appt Follow-Up:** Recall List, CRM Communication Log.
- **32 Telemedicine:** Virtual Waiting Room, Video Consultation UI, Chat Panel.
- **40 AI Prescription Capture:** Image Upload/Camera UI, AI Extraction Review Split-Screen.

SECTION 16: SAMPLE DATA STANDARDS

To ensure consistent UI testing and training, official sample data profiles must be used:

- **Patient:** John Doe (Male, 45), Amina Begum (Female, 32).
- **Doctor:** Dr. Sarah Smith (Cardiology), Dr. Ahmed Khan (Pathology).
- **Lab Order:** Complete Blood Count (CBC), Lipid Profile.
- **Admission:** General Ward, Bed A1.
- **Prescription:** Paracetamol 500mg (1-1-1), Amoxicillin.
- **Invoice:** Standard OPD Consultation + Lab Tests.
- **Report:** Normal CBC Report, Abnormal Lipid Profile (for flag testing).
- **Telemedicine Session:** Standard 15-minute follow-up.
- **AI Prescription:** Scanned handwritten image mapped to Paracetamol.

SECTION 17: MASTER MENU ARCHITECTURE

- **Host Menu:** Dashboard, Tenants, Global Settings, License Management, Host Audit.
 - **Tenant Menu:** Dashboard, Hospital Setup, Master Data, User Management, Reports.
 - **Department Menus:** Contextual. (e.g., "Laboratory" menu shows Sample Collection, Result Entry, Verification).
 - **Role Menus:** Dynamically generated based on Module 03 (Role Permissions).
 - **Portal Menus:** Home, My Appointments, My Reports, My Prescriptions, Profile.
 - **Mobile Menus:** Bottom tab bar: Home | Appointments | Records | More.
-

SECTION 18: DESIGN TOKENS & FUTURE Figma READINESS

To bridge the gap between UI design and development, ABSHealthcareLite will utilize Design Tokens:

- **Design Tokens:** Variables for colors (`--color-primary`), typography (`--font-size-h1`), and spacing (`--spacing-md`) stored in JSON/CSS variables.
- **Component Library:** Standardized UI components (Buttons, Inputs, Modals) ready for future React/Angular/Blazor componentization.
- **Theme Engine:** Allows tenants to inject their brand colors dynamically.
- **Dark Mode Readiness:** UI tokens support automatic inversion for a future Dark Mode release (critical for Radiology reading rooms).
- **Branding Engine:** Tenant logos automatically scale and fit into headers, reports, and patient portals.

SECTION 19: MOCKUP DEVELOPMENT ROADMAP

- **Phase 1 - Core Screens:** Login, Dashboard, Patient Registration, Master Data Grids.
- **Phase 2 - Clinical Screens:** Doctor Worklist, EMR Encounter, Prescription (CPOE).
- **Phase 3 - Diagnostic Screens:** Billing POS, Sample Collection, Result Entry, Verification.
- **Phase 4 - IPD Screens:** Visual Bed Board, Nursing Station, MAR, Discharge.
- **Phase 5 - Portal + Telemedicine:** Patient Mobile App screens, Video UI.
- **Phase 6 - AI Screens:** AI Prescription Capture review interfaces.

SECTION 20: APPENDIX A - STANDARD PAGE TEMPLATES

1. Standard List / Grid Template

[Logo] ABSHealthcareLite | [Branch] Main Hospital | [User] Dr. Smith | [Logout]

[Left Nav] | PAGE TITLE: Patient List

[Dashboard] |

[Patients] |

[Search Name/ID...] [Filter]

[+ Add Patient]

[Appts]		-----					
[Billing]		ID	Name	Gender	Phone	Action	
[Lab]		-----	-----	-----	-----	-----	
[Pharmacy]		P001	John Doe	M	555-0101	[View] [Edit]	
[Settings]		P002	Amina Begum	F	555-0102	[View] [Edit]	

		Showing 1-10 of 500 records				[< Prev] [Next >]	

2. Standard Entry / Form Template

[Left Nav]		PAGE TITLE: Add New Patient	

		DEMOGRAPHICS	
		First Name: [_____]	Last Name: [_____]
		DOB: [__/__/____]	Gender: [v Dropdown]
		CONTACT INFO	
		Phone: [_____]	Email: [_____]

		[Cancel]	[Save Patient]

3. Split-Screen Approval / AI Review Template

[Left Nav]		PAGE TITLE: AI Prescription Verification	

		[SOURCE IMAGE / DOCUMENT]	[EXTRACTED DATA (AI)]
		+-----+	Patient: [John Doe]
		(Scanned handwritten	
		prescription image	Medication:
		displayed here)	[Paracetamol 500mg] (98%
			Dose: [1-1-1] (95%
		+-----+	

|

|

[Reject AI] [Verify & Save]

SECTION 21: APPENDIX B - ABSHEALTHCARELITE DESIGN COMMANDMENTS

1. **Maximum information with minimum clicks:** Flatten navigation where possible.
 2. **Never hide critical patient information:** Allergies, blood type, and critical alerts must be universally visible in the patient header.
 3. **Safety before aesthetics:** Contrast and readability trump modern "minimalist" faded text.
 4. **Auditability before convenience:** Never allow a user to bypass a required audit reason for a deletion or modification.
 5. **Mobile readiness by default:** Design for the smallest screen first for all patient and doctor-facing apps.
 6. **Localization by default:** Never hardcode English text in the UI layer; always use resource keys.
 7. **AI recommendations never bypass human approval:** AI is an assistant, not a doctor. Human-in-the-loop is mandatory.
-

SECTION 22: CONCLUSION

The **ABSHealthcareLite UI/UX Blueprint** establishes a rigorous, standardized approach to product design. By adhering to these guidelines, the development and design teams will ensure that the platform remains intuitive, clinically safe, and highly efficient.

This design system guarantees that as ABSHealthcareLite scales—from ASP.NET WebForms to future MVC/Core APIs, and from desktop web to native mobile apps—the user experience will remain consistent, accessible, and fully compliant with the platform's multi-tenant, multilingual, and AI-governed architectural vision.

